STAGE 1: EXTERNAL DATA SOURCES

NSE & BSE Archives
2,639 equities listed
ISIN Registry (NSE ↔ BSE)
Historical OHLCV archives
Corporate action records
Batch / Bulk Download

Yahoo Finance API
1-Year OHLCV (daily close)
Fundamentals: P/E, ROE, P/B
Market cap & EV metrics
Dividend yield & EPS
REST / yfinance

Reserve Bank of India
Repo Rate: 6.50%
CPI Inflation: 5.10%
MPC policy decisions
Liquidity & FX data
RBI Open Data Portal

NSE Stream & News
India VIX (volatility index)
Nifty 50 live / EOD
Corporate announcements
Business news headlines
NSE Feed / RSS

Google Gemini API
LLM reasoning engine
Multi-step chain-of-thought
Structured JSON output
Gemini 1.5 Pro model
REST / google-generativeai

STAGE 2: NORMALIZATION & STORAGE

ISIN Resolver
NSE ↔ BSE ticker mapping
Deduplication by ISIN code
Exchange canonical form
Symbol normalization rules
ISIN → Canonical Ticker

Liquidity Funnel
2,639 → 504 liquid stocks
ADTV filter: ≥ ₹5 Lakh/day
Rolling 90-day average volume
Penny stock & halt exclusion
2,639 → 504 Liquid Universe

Factor Z-Scores
ROE, P/E, P/B cross-sectional
Price momentum (52-week)
Composite quality score
Sector-relative ranking
Z = (x − μ) / σ per sector

PostgreSQL 16
ACID transactions
SQLAlchemy 2.0 ORM
Alembic migrations
62 tables / 8 domains
Persistent data store

STAGE 3: AGENT PAYLOAD ASSEMBLY

Investor Mandate
Capital to invest & risk score (1–10)
Sector ban list & ESG prefs
Investment horizon & goals
User profile from DB

Macroeconomic Context
Repo Rate 6.50% → rate regime tag
CPI 5.10% → inflation tag
VIX regime: low / mid / high
Macro snapshot from DB

Pre-Screened Universe
504 verified liquid stocks
With factor scores & fundamentals
Sector tags & market-cap bands
Screened equity list

STAGE 4: LANGGRAPH MULTI-AGENT RESEARCH

Sequential Research Chain →

Macro Analyst
Rate & inflation regime analysis
Sector tailwind / headwind
→ Macro memo

Market Analyst
Technical trend & momentum
Sector rotation signals
→ Market memo

Fundamental Analyst
P/E, ROE, P/B, EPS scoring
Quality & value ranking
→ Fundamental memo

News Analyst
Sentiment & event scoring
ESG & risk flagging
→ News memo

Lead Strategist — synthesises all 4 memos → long list → **Adversarial Critic — challenges & refines** → **Vetted Candidates: 10–16 stocks**

STAGE 5: DETERMINISTIC MATH & DELIVERY

SciPy SLSQP Optimizer
Objective: maximise Sharpe ratio
Weight bounds 5%–25% per stock
Sector concentration cap 40%

Whole-Share Rounding
Integer lot allocation per security
Cash reserve ≥ ₹0 enforced
Pro-rata rounding algorithm

Audit & Transaction Levies
STT, Stamp Duty, Exchange & GST
Fiduciary output: net cost & rationale
ACID commit to PostgreSQL DB